Module: 7 Reliability

- Basic concepts- Hazard function- Reliabilities of series and parallel systems- System Reliability - Maintainability-Preventive and repair maintenance- Availability.

Reliability – The probability that a failure may not occur in a given time interval.

Definition: Reliability of unit (product) is the probability that the unit performs its intended function adequately for a given period of time under the stated operating conditions or environment.

By a unit – an element, a system or a part of a system, or the like.

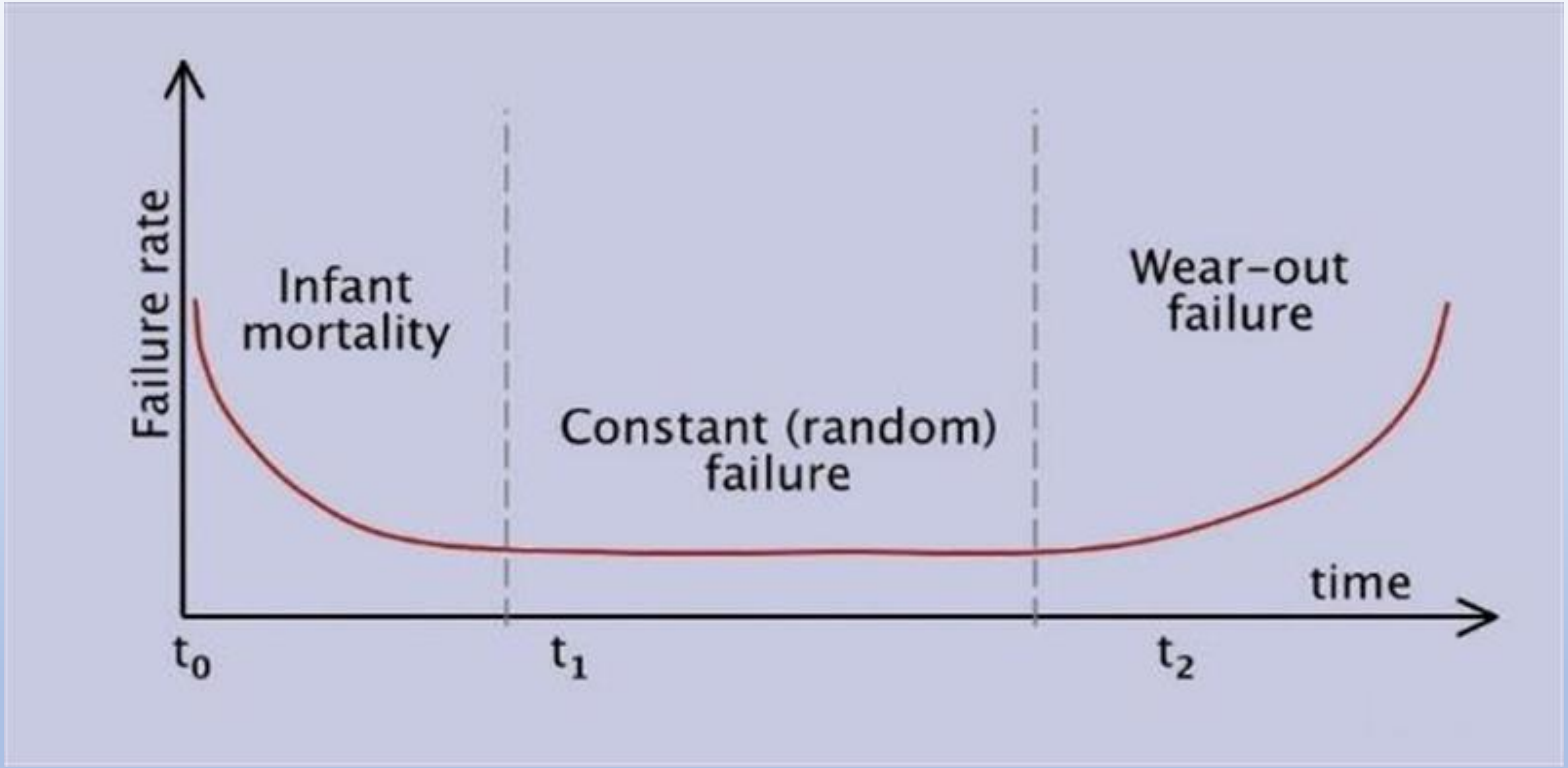
The definition of Reliability stresses four elements namely:

1. Probability
2. Intended function
3. Time
4. Operating conditions

If $\boldsymbol{T}$ is the time till the failure of the unit (*a random variable*) occurs, then the probability that it will not fail in a given environment before time $\boldsymbol{t}$ (*or its reliability*) is

$$R(t) = P(T > t) \quad ; \quad t > 0$$

Thus the reliability is always a function of time.


Failure rate
Infant mortality
Constant (random) failure
Wear-out failure
time
t_0
t_1
t_2

$$R(t) = P(T > t) \quad \text{or} \quad 1 - P(T \leq t)$$

$$= 1 - F(t)$$

$$F(t) = \int_0^t f(t) \; dt$$

Thus, $$R(t) = 1 - \int_0^t f(t)\, dt \; = \; \int_t^{\infty} f(t) \; dt$$

Since $F(0) = 0$ and $F(\infty) = 1$ by the property of *c.d.f.*

$R(0) = 1$ and $R(\infty) = 0$; *i.e.*, $0 \leq R(t) \leq 1$

Also, since $\frac{d}{dt}\,F(t) = f(t) \quad \Rightarrow \quad f(t) = -\,\frac{d\,R(t)}{dt}$

Now the conditional probability of failure in the time interval $(\boldsymbol{t},\ \boldsymbol{t} + \Delta\boldsymbol{t})$, Given that the component has survived upto time ***t,***

$$P\{t \le T \le t + \Delta t\ /\ T \ge t\ \} = \frac{f(t)\Delta t}{1-F(t)} \ = \ \frac{f(t)\Delta t}{R(t)}$$

The conditional probability of failure per unit time is called as the ***instantaneous failure rate*** or ***hazard rate*** of the component, denoted by $\boldsymbol{\lambda}(\boldsymbol{t})$.

$$\lambda(t) = \frac{f(t)}{R(t)}$$

$$-\frac{R'(t)}{R(t)} = \lambda(t)$$

Integrating on both sides w.r.t to t between 0 and t ,

$$\int_0^t \frac{R'(t)}{R(t)}\, dt = -\int_0^t \lambda(t)dt$$

$$\{\log R(t)\}_0^t = -\int_0^t \lambda(t)dt$$

$$\log R(t) - \log R(0) = -\int_0^t \lambda(t)dt$$

$$\log_e R(t) = -\int_0^t \lambda(t)dt$$

$$\boldsymbol{R(t) = e^{-\int_0^t \lambda(t)dt}}$$

$$\therefore \qquad \boldsymbol{f(t) = \lambda(t)\, e^{-\int_0^t \lambda(t)dt}}$$

Mean Time to Failure (MTTF)

$$MTTF \;=\; E(T) \;= \int_0^{\infty} t\, f(t)\, dt \quad = \; -\int_0^{\infty} t\, R'(t)\, dt$$

$$= \; [t\, R(t)]_0^{\infty} + \int_0^{\infty} R(t)\, dt$$

(On integration by parts)

$$MTTF \;=\; \int_0^{\infty} R(t)\, dt$$

$$Var\,(t) = \int_0^{\infty} t^2 f(t)\, dt \;-\; (MTTF)^2$$

Conditional Reliability:

Reliability of a component or system following a wear in period (burn-in period) or after a warranty period :

$$\mathrm{R}(t/\,T_0) = \; P\{T > T_0 + \; t/T > T_0\}$$

$$= \; \frac{P\{T > T_0 + t\,\}}{P\{T > T_0\}} \; = \frac{\mathrm{R}(\,T_0 + t)}{\mathrm{R}(\,T_0)}$$

$$= \; \boldsymbol{e}^{-\int_{T_0}^{T_0 + t} \boldsymbol{\lambda}(\boldsymbol{t})\boldsymbol{dt}}$$

Exponential Distribution Function:

$$f(t) = \lambda\, e^{-\lambda t} \;\; ; \;\; t \geq 0$$

$$R(t) = \int_t^{\infty} f(t)\; dt = \int_t^{\infty} \lambda\, e^{-\lambda t}\; dt = \; e^{-\lambda t}$$

$$\lambda(t) = \frac{f(t)}{R(t)} \; = \; \frac{\lambda\, e^{-\lambda t}}{e^{-\lambda t}} \; = \lambda$$

i.e.., when the failure time is exponentially distributed with the parameter $\boldsymbol{\lambda}$, the failure rate at any time is constant, equal to $\boldsymbol{\lambda}$.

Conversely, when $\boldsymbol{\lambda(t)}$ = a constant $\boldsymbol{\lambda}$,

$$\boldsymbol{f(t) = \lambda\, e^{-\int_0^t \lambda\, dt} = \; \lambda\, e^{-\lambda t} \quad , t \geq 0}$$

Due to this property, the exponential distribution is often referred to as ***constant failure rate distribution*** in reliability contexts.

Weibull Distribution Function

$$f(t) = \alpha\beta\, t^{\beta-1}\, e^{-\alpha t^{\beta}} \quad , \quad t \geq 0$$

By putting, $\alpha = \frac{1}{\theta^{\beta}}$, β - shape parameter , $\theta -$ scale parameter

$$f(t) = \frac{\beta}{\theta}\,(\frac{t}{\theta})^{\beta-1}\; e^{-(\frac{t}{\theta})^{\beta}} \;,\; \theta > 0,\;\; \beta > 0, t \geq 0$$

$$R(t) = \int_t^{\infty} \frac{\beta}{\theta}\,(\frac{t}{\theta})^{\beta-1}\; e^{-(\frac{t}{\theta})^{\beta}}\; dt \quad = \; e^{-(\frac{t}{\theta})^{\beta}}$$

$$\lambda(t) = \frac{f(t)}{R(t)} \; = \frac{\beta}{\theta}\,(\frac{t}{\theta})^{\beta-1}$$

$$MTTF = E(T) = \theta\,\Gamma\left(1 + \frac{1}{\beta}\right) \;;$$

$$\mathrm{Var}\ (\mathrm{T}) = \theta^2 \left\{\Gamma\left(1 + \frac{2}{\beta}\right) - \left[\Gamma\left(1 + \frac{1}{\beta}\right)\right]^2\right\}$$

The failure density function of random variable T is given by

$$f(t) = \begin{cases} 0.012e^{-0.012t}, & t \geq 0 \\ 0, & otherwise \end{cases}$$

Calculate

i. Reliability of the component.
ii. Reliability of the component for a 100-hour mission time.
iii. Mean time to failure (MTTF).
iv. What is the life of the component if a reliability of 0.96 is desired?

Given failure density function of the random variable T is

$$f(t) = \begin{cases} 0.012e^{-0.012t}, & t \geq 0 \\ 0, & otherwise \end{cases}$$

i. The reliability is

$$R(t) = \int_t^{\infty} f(t)\,dt = \int_t^{\infty} 0.012e^{-0.012t}\,dt = 0.012\left(\frac{e^{-0.012t}}{-0.012}\right)_t^{\infty} = e^{-0.012t}$$

ii. The reliability of the component for a 100 hour mission time is when $t = 100$

$$R(100) = e^{-0.012t} = e^{-0.012\times 100} = e^{-1.2} = 0.3012$$

iii. $MTTF = \int_0^\infty R(t)\ dt = \int_0^\infty e^{-0.012t}\ dt = \left(\frac{e^{-0.012t}}{-0.012}\right)_0^\infty = -\frac{1}{0.012}(0 - 1) = 83.33$

iv. Let t_{md} denotes the median time to failure

$$R(t_{md}) = P(T > t_{md}) = 0.5$$

$$R(t_{md}) = \int_{t_{md}}^{\infty} f(t)\, dt = 0.5$$

$$e^{-0.012 t_{md}} = 0.96$$

$$-0.012\ t_{md} = \ln\ 0.96$$

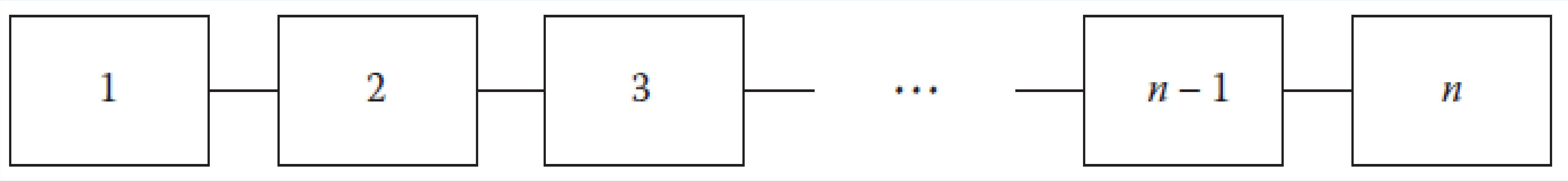
$$-0.012\ t_{md} = -0.0408$$

$$t_{md} = 3.4\ \text{hrs}$$

System Reliability

- A system is generally understood as a set of components assembled to perform a certain function.
- To evaluate the reliability of a complex system, we may apply a particular-failure law to the entire system.
- But it will be proper if we determine an appropriate reliability model for each component and then compute the reliability of the system by applying the relevant rules of probability according to the configuration of the components within the system.

Series Configuration: If the Components are connected in a series:



All components must function for the system to function.

Let $R_1(t)$, $R_2(t)$,…., $R_n(t)$ and $R_s(t)$ be the reliabilities of the components 1 and 2 ….. n.

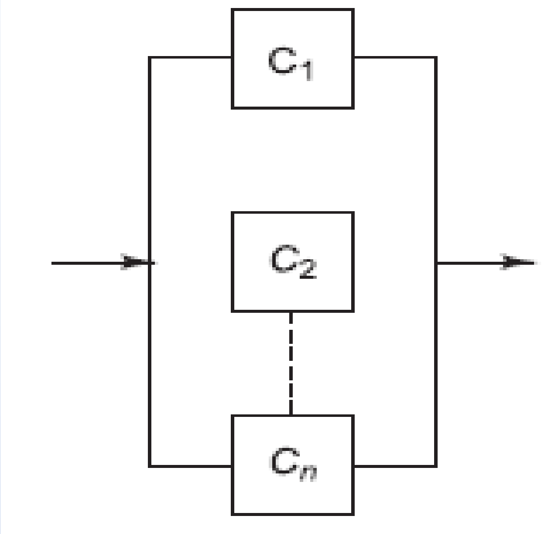
Then $$R_s(t) = R_1(t) \times R_2(t) \times \cdots \times R_n(t)$$

$$= \prod_{i=1}^{n} R_i(t)$$

Also, $R_s(t) \le min\{R_1(t), R_2(t), \dots, R_n(t)\}$

i.e., the system reliability will not be greater than the smallest of the component reliabilities.

Parallel Configuration: Parallel or redundant configuration is one in which the components of the system are connected in parallel,



All components must fail for the system to fail. ***i.e.,*** If one or more components function, the system continues to function.

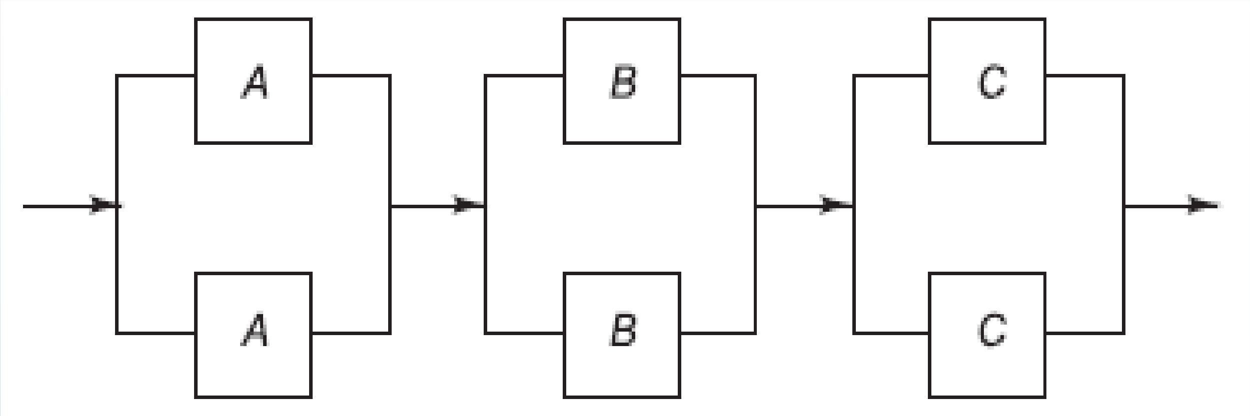
Let $R_1(t)$, $R_2(t)$,…., $R_n(t)$ and $R_P(t)$ be the reliabilities of the components 1 and 2 …. . . n.

$$R_P(t) = 1 - (1 - R_1) \times (1 - R_2) \times \cdots \times (1 - R_n)$$

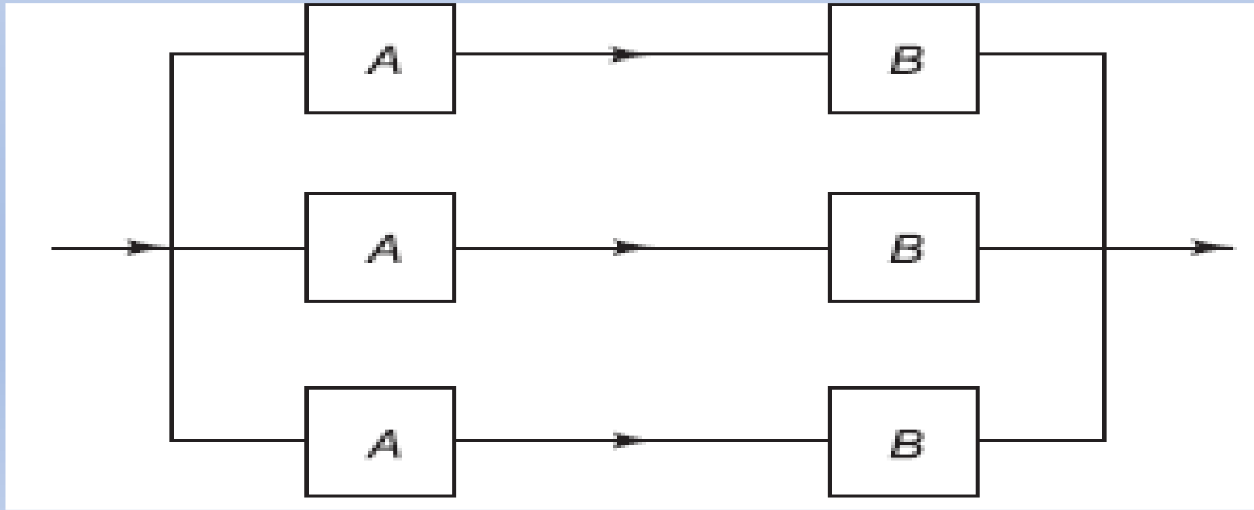
$$= 1 - \prod_{i=1}^{n}[1 - R_i(t)]$$

$$R_P(t) \geq Max\,\{R_1(t), R_2(t), \ldots, R_n(t)\}$$

Parallel – Series (low-level redundancy)



Series – Parallel (high-level redundancy)



Maintainability:

A measure of how fast a system may be repaired following failure is known as maintainability. Repairs require different lengths of time and even the time to platform a given repair is uncertain, because circumstances skill level, experience of maintenance personnel and such other factors vary. Hence the time T required to repair a failed system is a continuous random variable.

Maintainability is mathematically defined ass the cumulative distribution function of the R.V.T representing the time to repair and denoted as $M(t)$

$$M(t) = P(T \leq t) = \int_0^t m(t)\ dt$$

Where $m(t)$ is the pdf of T

Mean time to repair:

The expected value of repair time T is called the mean time to repair (MTTR) and is given by

$$MTTR = E(T) = \int_0^t t \times m(t)\ dt$$

If the conditional probability that the system will be repaired between t and $t + \Delta t$, given that it has failed at t and the repair starts immediately, is $\mu(t)\,\Delta t$, then $\mu(t)$ is called the instantaneous repair rate or simply repair rate and denotes the number of repairs in unit time.

i.e., $\mu(t)\,\Delta t = \frac{P(t \le T \le t + \Delta t)}{(P(T > t)}$

$$\mu(t) = \frac{\left(m(t)\right)}{1 - M(t)}$$

$$m(t) = \frac{d}{dt} M(t)\,,$$

$$\mu(t) = \frac{M'(t)}{1 - M(t)}$$

$$M(t) = 1 - e^{-\int_0^t \mu(t)\,dt}$$

$$m'(t) = \mu(t) e^{-\int_0^t \mu(t)\,dt}$$

The time to repair a power generator is best described by its pdf

$$m(t) = \frac{t^2}{333}, \qquad 1 \le t \le 10 \quad hours$$

i. Find the probability that a repair will be completed in 6 hours.
ii. What is the MTTR
iii. Find the repair rate.

$i.\ \ P(T < 6) = P(1 \leq T < 6),$ where T is the time to repair

$$P(T < 6) = \int_1^6 m(t)\,dt = \int_1^6 \frac{t^2}{333}\ dt = 0.2152$$

ii. $MTTR = \int_0^t t \times m(t)\ dt = \int_1^{10} \frac{t^3}{333}\ dt = \left(\frac{t^4}{4 \times 333}\right)_1^{10} = 7.5$ hours

iii. Repair rate $\mu(t) = \frac{m(t)}{1-M(t)} = \frac{\frac{t^2}{333}}{\int_t^{10} \frac{t^2}{333}\ dt} = \frac{\frac{t^2}{333}}{\frac{1}{999}(10^3 - t^3)}$

$= \frac{3t^2}{1000 - t^3}$ per hour

Practice Problem:

The density function of the time to failure in years of the gizmos (for use on widgets) manufactured by a certain company is given by

$$f(t) = \frac{200}{(t+10)^3} \quad , \quad t \geq 0.$$

(i) Derive the reliability function and determine the reliability for the first year of operation.

(ii) Compute the MTTF.

(iii) What is the design life for a reliability 0.95?